

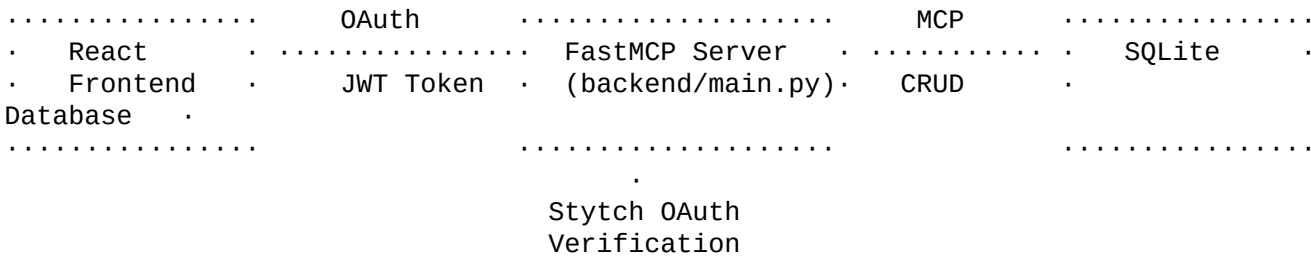
Advanced MCP Server With Auth

A **Model Context Protocol (MCP)** server with **OAuth authentication** (via Stytych), user-scoped note management, and a React frontend client.

Features

- **OAuth Authentication** • Stytych-based user auth with JWT token validation
- **Note Management** • Create, read, and list notes per authenticated user
- **SQLite Storage** • Lightweight persistent storage with SQLAlchemy
- **React Frontend** • Client UI for interacting with the MCP server
- **User Scoping** • Notes are isolated per authenticated user

Architecture



File Structure

```
AdvancedMCPServerWithAuth/
... backend/
  ... main.py          # FastMCP server with auth and note tools
  ... database.py      # NoteRepository with SQLAlchemy + SQLite
  ... pyproject.toml   # Backend dependencies
... frontend/
  ... src/             # React app source
  ... package.json     # Frontend dependencies
```

Key Backend Files

| File | Purpose | |----|-----| | backend/main.py | FastMCP server definition, OAuth middleware, MCP tool endpoints | | backend/database.py | NoteRepository class · SQLAlchemy models, CRUD operations |

MCP Tools

| Tool | Description | |----|-----| | add_note(content) | Create a new note for the authenticated user | | read_notes() | List all notes for the authenticated user |

Getting Started

Prerequisites

- Python 3.10+
- Node.js 18+
- Styth account (for OAuth)

Backend Setup

```
cd backend
uv sync
```

Create `.env`:

```
STYTCH_PROJECT_ID=...
STYTCH_SECRET=...
```

```
uv run python main.py
```

Frontend Setup

```
cd frontend
npm install
npm run dev
```

· Logic Flow

- 1 **Authentication** · User authenticates via Styth OAuth
 - 2 **JWT Validation** · Backend validates JWT token on each request
 - 3 **User Scoping** · User ID extracted from token scopes all operations
 - 4 **Tool Execution** · MCP client calls `add_note` or `read_notes`
 - 5 **Database** · SQLAlchemy persists notes to SQLite with user association
 - 6 **Response** · Results returned via MCP protocol
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· Dependencies

Backend: fastmcp, python-jose, sqlalchemy, styth, python-dotenv **Frontend:** React, TypeScript
(see frontend/package.json)

· License

Educational project · use freely for learning and reference.